PROJECT JOURNEY

From One Forecast to a Working Model

A concise record of how a question about the Netherlands became an auditable, live World Cup prediction system.

SCOPE	STATUS	PREPARED
2026 World Cup model v2.14	**Research prototype**	**13 October 2026**

ORIGIN	**The project began by asking how Joachim Klement could forecast a World Cup winner, then deliberately built an independent model instead of copying a proprietary one.**

The spark

The first question was simple: what sits behind a bold call such as 'the Netherlands will win'? Public material described an econometric approach using football strength, GDP per capita, population, climate, host effects and randomness, but not the coefficients, raw dataset or full simulation code. That gap defined the project. We would use the idea as a challenge case, then build a model whose inputs and decisions could be explained.

The rules we set early

- Football evidence first. GDP and population could remain as low-weight context, but Elo, squads, form, tactics and availability had to drive the result.
- No fake certainty. Exact scores could exist inside goal simulations, but the public product would emphasize 90-minute direction, knockout advancement, confidence and Pass.
- Missing players are replacement problems, not fame problems: impact equals the starter's value minus the replacement's value.
- Every tournament path must be simulated match by match through the real groups and bracket; a simple ranking table is not a World Cup forecast.
- The model must move. Results, weather and confirmed lineups should update the next prediction without rewriting history.

Building the foundation

49,296	1,246	128	104
historical internationals	2026 squad players	open-data WC matches	2026 fixtures

Historical results produced a local Elo and recent-form timeline. StatsBomb's 2018/2022 open events supplied tactical prototypes. OpenFootball supplied the tournament structure. Public squad tables seeded a custom Player Impact Rating, while World Bank and FIFA snapshots supported research and sanity checks. The first complete system could already run groups, third-place qualification and knockouts with Monte Carlo randomness.

The first model thesis

Team strength ✦ current squad ✦ matchup context ✦ bounded randomness -• match probabilities -• full tournament simulation

The build: v1 to v2.7

Each version answered a specific criticism raised during the build. The version number therefore records a decision, not just more code.

Version	Question solved	What changed
v1	Working MVP	Combined local Elo, recent form, ceiling, host effect, tactical priors and availability; simulated the tournament rather than ranking teams once.
v2	Layered scoring	Normalized the model into 0-100 components. An initial 15% manager weight was later judged too subjective.
v2.1	Player impact	Rated 1,246 players from ability, club level, national role, tactical fit, sharpness and experience; squad depth and replacement gap became explicit.
v2.2	Draw repair	Made the draw a first-class 1/X/2 outcome and recalibrated low-scoring match behavior.
v2.3	Bias control	Reduced manager quality from 15% to 3%, neutralized unknowns, added explicit knockout randomness and ran 50,000 tournament simulations.
v2.4-2.5	Live plumbing	Connected API-Football and weather, added request-budget protection, and created structured tables for fixtures, lineups, predictions and post-match results.
v2.6	Model bake-off	Compared Elo, Poisson-lite, logistic regression, random forest, gradient boosting and blends on chronological train/calibration/test windows.
v2.7	Integrated engine	Trained the selected match model through 2 June 2026 and connected calibrated probabilities to the tournament simulator.

Two models working together

Layer	Selected weights
Tournament/team profile	Elo 30% | squad 32% | recent form 10% | tactics 10% | availability 7% | market 5% | ceiling 3% | manager 3%
Calibrated 1/X/2 ensemble	Gradient boosting 50% | Elo formula 30% | logistic regression 20% | Poisson-lite 0% after the bake-off

Why this mattered: the team-profile layer expresses World Cup-specific strength; the trained ensemble converts matchup features into calibrated 90-minute probabilities. The bracket simulator then resolves extra time, penalties and path dependence.

The tournament system: v2.8 to v2.14

Version	Product step	What changed
v2.8	Public language	Introduced Strong Pick, Model Lean and Pass so uncertainty became part of the product.
v2.9	Matchday reports	Combined fixture state, weather, lineups, 90-minute read and full-match advancement; exact scores were removed from public reports.
v2.10	Honest validation	Scored only predictions archived before kickoff. Late backfills remained visible but were excluded.
v2.11	Stricter gate	Softened extreme probabilities, added a small draw uplift and required stronger agreement and data quality.
v2.12	Confirmed XIs	Required 11 starters for both teams, at least 90% identity matching and a five-point cap on lineup-driven probability movement.
v2.13	Live tournament	Locked completed results and repeatedly simulated only the remaining bracket, allowing title probabilities to evolve.
v2.14	Full audit	Produced the pre-final fixture ledger, gate breakdown, probability diagnostics and honest business verdict.

What the World Cup test actually showed

65.38%	83.33%	82 / 103
group 90-min accuracy	knockout advancement	valid pre-final calls

The primary live scorecard contained 82 frozen pre-kickoff calls before the final. Group matches were graded on the best 90-minute 1/X/2 read (34/52). Knockouts were graded on the team advancing after extra time or penalties (25/30). Combined, that was 59/82, or 71.95%, but it is a hybrid product metric and should always be explained that way. A separate static pre-tournament v2.3 sheet covered all 72 group matches and scored 48/72 (66.67%).

What worked, and what did not

Read	Finding
Worked	Knockout advancement was the strongest product; quarter-finals were 4/4. Model Lean was the most useful live gate, and the audit trail improved as the tournament progressed.
Weakness	Draw recall was 5/16 (31.25%). Early operational coverage was incomplete, Strong Pick had too little live evidence, and the third-place match exposed under-modelled goal variance.
Lesson	Missed chances can explain process, but they cannot turn a wrong call into a correct one. Event xG and post-shot xG are needed to measure bad luck instead of arguing it.

Where the project arrived

The result is a credible research-stage prediction system: reproducible datasets, a calibrated ML ensemble, real bracket simulation, weather and lineup overlays, confidence gating, immutable validation logic, and a complete tournament audit. It is not yet a proven commercial edge. The next serious milestone is automated T-minus snapshots for every fixture, licensed event/availability data, a dedicated draw model, and benchmarking against no-vig closing markets over several competitions.

The project began with one man's Netherlands forecast. It ended with something more valuable: a model that can explain what it knows, admit what it does not, and preserve the evidence either way.